

BIO401 Best File For MID Term Preparation

Statistics is a field of study that is concerned with:

- The collection, organization, summarization, and analysis of data.
- The drawing of inferences about a body of data, when only a part of data is observed.

Biostatistic is

- The use of statistical tools for the data that is derived from biological sciences and medicine

Applications of Biostatistics:

- Public Health
- Pharmacology
- Epidemiology
- Medicine
- Genetics
- Genomics
- Proteomics
- Bioinformatics etc etc

Population: In Statistics it is Defined as “an aggregate of entities for which we have an interest at a particular time.” It can consists of Animals, machines, places or cells etc.

Example: If we are interested in knowing the weights of all the children enrolled in elementary schools of the Bahawalpur District. Then our population will be all the the children enrolled in elementary schools of the Bahawalpur District.

Types of Population

1. Finite

If a population of values consists of a fixed number of these values, the population is said to be finite

The finite population is also known as a countable population in which the population can be counted. In other words, it is defined as the population of all the individuals or objects that are finite. For statistical analysis, the finite population is more advantageous than the infinite population. **Examples** of finite populations are employees of a company, potential consumer in a market.

2. Infinite

If a population consists of an endless succession of values, the population is infinite one.

The infinite population is also known as an uncountable population in which the counting of units in the population is not possible. **Example** of an infinite population is the number of germs in the patient's body is uncountable.

Census

- A Census is a survey conducted on a full set of observations belonging to a given population.

Population Parameter:

It is any summary number, like an average or percentage, that describes the entire population.

- These are the true values, which are usually unknown.

These values are denoted by Greek letters. e.g.: Population Mean μ (Mu)

, Population Proportion π (Pi)

Sample Statistic

Any summary number, like an average or percentage, that describes the sample.

- Sample statistics are random variables.
- These are the values which are usually denoted by Latin letters
- Sample Standard Deviation (s)
- Sample Proportion (p)

Sample is a representative part of a Population.

A **sample survey** is a study that obtains data from a subset of a population of our interest, in order to estimate population attributes.

Characteristics of a good sample

1. Representative.
2. It captures most of the variation in the population.
3. Obtained by using proper sampling methodology.
4. Focus on objective
5. Informative with minimum use of resources.

DATA:

THE raw material of statistics is called data. It is the collection of facts and figures. Which can occur as process of measurement or counting.

For example:

- When a hospital administrator counts the number of patients (counting).
- When a nurse weighs a patient (measurement)

Sources of Data

We search for suitable data to serve as the raw material for our investigation. Such data are available from one or more of the following sources:

1- Routinely kept records.

For example: - Hospital medical records contain immense amounts of information on patients.

-Hospital accounting records contain a wealth of data on the facility's business activities. **2- External sources.**

The data needed to answer a question may already exist in the form of published reports, commercially available data banks, or the research literature, i.e. someone else has already asked the same question.

3- Surveys:

The source may be a survey, if the data needed is about answering certain questions.

For example: If the administrator of a clinic wishes to obtain information regarding the mode of transportation used by patients to visit the clinic, then a survey may be conducted among patients to obtain this information.

4- Experiments.

Frequently the data needed to answer a question are available only as the result of an experiment.

For example: If a nurse wishes to know which of several strategies is best for maximizing patient compliance, she might conduct an experiment in which the different strategies of motivating compliance are tried with different patients.

Types of Data

Primary data refers to the first hand data gathered by the researcher himself. E.g: obtaining vitals of the patient directly by observing the patient(respondent)

. Primary data is the kind of data that is collected directly from the data source without going through

any existing sources. It is mostly collected specially for a research project and may be shared publicly to be used for another research.

Primary data is often reliable, authentic, and objective in as much as it was collected with the purpose of addressing a particular research problem. It is noteworthy that primary data is not commonly collected because of the high cost of implementation.

Secondary data means data collected by someone else earlier. e.g obtaining vitals of the patient from the available medical records.

Secondary data is the data that has been collected in the past by someone else but made available others to use. They are usually once primary data but become secondary when used by a third party.

Secondary data are usually easily accessible to researchers and individuals because they are mostly shared publicly. This, however, means that the data are usually general and not tailored specifically to meet the researcher's needs as primary data does.

Types of Statistics

1. Descriptive Statistics

These are the statistical methodologies which are used to Organize and Summarize data. This summarization can be in numeric as well as in graphical form. We use different methodologies within them. And these descriptive methodologies form the basis of any

quantitative analysis of data Descriptive analysis is also known as Exploratory Data Analysis (EDA)

2. Inferential Statistics

These are the methods which we used to talk about the population from The information that comes out of the sample. OR. These are statistical methodologies using which we reach a conclusion about a population on the basis of the information contained in the sample.

Inference

Use a random sample to learn something about a larger population

Variable

It is a characteristic that takes on different values in different persons, places, or things.

For example:

- heart rate,
- the heights of adult males,
- the weights of preschool children,
- the ages of patients seen in a dental clinic.

Types of Variable Quantitative

Variable:

it is the one that can be measured and expressed numerically.

Discrete variable: it is the ones that have gap or jumps in the possible values. These gaps indicate the absence of values between particular possible values of our variable. It is also known as countable values. E.g: no of pages in book, teeth per child in a school, children in a family.

Continuous variable: It is the ones that take on each and every possible value within an interval. Don't possess gaps in their possible values. E.g; height of a person, weight of a person.

Qualitative variable: It is the one that can not be measure in a numerical form. These are non-numeric variable where each possible value is a category of a variable that's why they are also known as categorical variable or attribute. E.g; brand of PC, hair color, marital status, category of different types of animals etc.

Measurement

It is Defined as the assignment of numbers to objects or events according to a set of rule.

Properties of measurement scales:

Each scale of measurement satisfies one or more of the following properties of measurement.

Identity. Each value on the measurement scale has a unique meaning.

□ **Magnitude.** Values on the measurement scale have an ordered relationship to one another. That is, some values are larger and some are smaller.

□ **Equal intervals.** Scale units along the scale are equal to one another. This means, for example, that the difference between 1 and 2 would be equal to the difference between 19 and 20.

□ **A minimum value of zero.** The scale has a true zero point, below which no values exist.

Types of Measurement Scales:

There are four types of measurement scales • Nominal • Ordinal • Interval • Ratio

Location parameter

The mean and median are special cases of a family of parameters known as location parameter. **Measures of position**

Other location parameters which helps us to determine the position of a particular observations in a data are called measures of position.

Quantile

These measures of position when help us to divide sample data into **n** equal part or divide the probability distribution into contiguous intervals with equal probabilities are called quantiles.

Types of quantiles

Quartile: In an array quartiles are 3 cut points which divide the values into 4 equal parts. Denoted by Q_k (k is a subscript).

Decile

In an array the Deciles are 9 values/cut-points that divides the values into 10 equal parts.

Denoted by D_k

Percentile

In an array the Percentiles are 99 values/cut-points that divides the values into 100 equal parts.

Denoted by P_k

2. **Random sample or probability sample:** The selection of units in the sample from a population is governed by the laws of chance or probability. The probability of selection of a unit can be equal as well as unequal.
3. **Non-random sample or purposive sample:** The selection of units in the sample from the population is not governed by the probability laws. For example, the units are selected on the basis of the personal judgment of the surveyor. The persons volunteering to take some medical test or to drink a new type of coffee also constitute the sample on non-random laws.

Probability Sampling

This Sampling technique uses randomization to make sure that every element of the population gets an equal chance to be part of the selected sample. It's alternatively known as random sampling. Such sampling techniques can be classified in the following way.

- Simple random sampling • Stratified sampling • Systematic sampling • Cluster (area) sampling • Multistage sampling

Simple Random Sampling: Every element has an equal chance of getting selected to be the part sample. It is used when we don't have any kind of prior information about the target population.

For example: Random selection of 20 students from class of 50 student. Each student has equal chance of getting selected. Here probability of selection is $1/50$

Stratified Sampling: This technique divides the elements of the population into small subgroups (strata) based on the similarity in such a way that the elements within the group are homogeneous and heterogeneous among the other subgroups formed. And then the elements are randomly selected from each of these strata. We need to have prior information about the population to create subgroups.

Cluster Sampling Our entire population is divided into clusters or sections and then the clusters are randomly selected. All the elements of the cluster are used for sampling. Clusters are identified using details such as age, sex, location etc. Cluster sampling can be done in following ways:

Single Stage Cluster Sampling Entire cluster is selected randomly for sampling.

Two Stage Cluster Sampling Here first we randomly select clusters and then from those selected clusters we randomly select elements for sampling

Systematic Clustering: Here the selection of elements is systematic and not random except the first element. Elements of a sample are chosen at regular intervals of population. All the elements are put together in a sequence first where each element has the equal chance of being selected.

Multi-Stage Sampling It is the combination of one or more methods described above. Population is divided into multiple clusters and then these clusters are further divided and grouped into various sub groups (strata) based on similarity. One or more clusters can be randomly selected from each stratum. This process continues until the cluster can't be divided anymore. For *example* country can be divided into states, cities, urban and rural and all the areas with similar characteristics can be merged together to form a strata.

Non-Probability Sampling Methods

It does not rely on randomization. This technique is more reliant on the researcher's ability to select elements for a sample. Outcome of sampling might be biased and makes difficult for all the elements of population to be part of the sample equally. This type of sampling is also known as nonrandom sampling.

• Convenience Sampling • Purposive Sampling • Quota Sampling • Referral /Snowball Sampling
Convenience Sampling: Here the samples are selected based on the availability. This method is used when the availability of sample is rare and also costly. So based on the convenience samples are selected. For *example*: Researchers prefer this during the initial stages of survey research, as it's quick and easy to deliver results.

Purposive Sampling This is based on the intention or the purpose of study. Only those elements will be selected from the population which suits the best for the purpose of our study. For Example: If we want to understand the thought process of the people who are interested in pursuing master's degree then the selection criteria would be "Are you interested for Masters in..?" All the people who respond with a "No" will be excluded from our sample.

Quota Sampling This type of sampling depends of some pre-set standard. It selects the representative sample from the population. Proportion of characteristics/ trait in sample should be same as population. Elements are selected until exact proportions of certain types of data is obtained or sufficient data in different categories is collected

For example: If our population has 45% females and 55% males then our sample should reflect the same percentage of males and females.

Referral /Snowball Sampling This technique is used in the situations where the population is completely unknown and rare. Therefore we will take the help from the first element which we select for the population and ask him to recommend other elements who will fit the description

of the sample needed. So this referral technique goes on, increasing the size of population like a snowball.

For *example*: It's used in situations of highly sensitive topics like HIV Aids where people will not openly discuss and participate in surveys to share information about HIV Aids. Not all the victims will respond to the questions asked so researchers can contact people they know or volunteers to get in touch with the victims and collect information

Descriptive Statistics is performed using *Tabular, Graphical, and Numerical methods*.

Tabular Methods are used to summarize the data in table form. It is a systematic organization of information in grid row and columnar structure. The most frequently used tabular format for data summarization is Frequency table and Cross-tabulation.

Graphical Methods are a visual way of presenting data using charts and graphs. The visuals make the data intuitive and self-understandable. The most frequently used visual representation of data are Bar Plot, Histogram, Pareto Chart, Box Plot, Pie Chart, Line Plot, and Scatter Plot.

Ordered array

It is the listing of the values of a collection(either sample or population) in order of magnitude from the smallest value to largest value.

Statistical Distribution

It is the distribution of statics across an infinite number of samples

Frequency Distribution

It is the organization of Raw data into tabular form using two important components of frequency distribution.

- Class. And Class frequency

Class is a quantitative or qualitative variable whereas **class frequency** is the number of data values in a specific class.

Class is a range for the values of a variable. **Frequency** is the number of observations associated with a class. **Relative Frequency** is the proportion of observations (frequency) associated with a class

Frequency distribution

• A table that organizes data values into classes or intervals along with number of values that fall in each class (frequency, f).

Types of frequency distribution

Categorical frequency Distribution : is used for qualitative data {Nominal & Ordinal}

Ungrouped frequency distribution: is used for quantitative data { Interval & Ratio}

for data sets with few different values, each value is in its own class

Grouped frequency distribution: is used for quantitative data. It shows the frequency of an item in each separate data value rather than groups of data values.

for data sets with many different values, which are grouped together in the classes.

Relative frequency

- Shows the portion or percentage of the data that falls in a particular class.

OR

A relative frequency is the ratio (fraction or proportion) of the number of times a value of the data occurs in the set of all outcomes to the total number of outcomes.

To convert a frequency into a proportion or relative frequency, we should divide the frequency for each class by the total of the frequencies. The sum of the relative frequencies will always be 1.

$R.F = \text{frequency of the class} / \text{total}$

Cumulative relative frequency

It is the accumulation of the previous relative frequencies. To find the cumulative relative frequencies, add all the previous relative frequencies to the relative frequency for the current row.

Cumulative relative frequency = sum of previous relative frequencies + current class frequency

Contingency tables

Contingency tables (also called crosstabs or two-way tables) are used in statistics to summarize the relationship between several categorical variables. A contingency table is a special type of frequency distribution table, where two variables are shown simultaneously. The entries in the cells of a two-way table can be frequency counts or relative frequencies (just like a one-way table).

Frequency Histograms

A graphical display of distribution of frequencies is known as frequency histogram.

A bar graph that represents the frequency distribution.

- The horizontal scale is quantitative and measures the data values.
- The vertical scale measures the frequencies of the classes.
- Consecutive bars must touch.

HISTOGRAM

A histogram consists of a set of adjacent rectangles whose bases are marked off by class boundaries along the X-axis, and whose heights are proportional to the frequencies associated with the respective Classes.

FREQUENCY POLYGON:

A frequency polygon is obtained by plotting the class frequencies against the mid-points of the classes, and connecting the points so obtained by straight line segments.

FREQUENCY CURVE: When the frequency polygon is smoothed, we obtain what may be called the frequency curve.

Box and whisker plot

It shows five number summary of the set of data: *minimum, lower quartile, median, upper quartile, maximum.*

A box plot (also called a box and whisker plot) shows data using the middle value of the data and the quartiles, or 25% divisions of the data.

Construction:

Step 1: Arrange the data in ascending order.

Step 2: Find the median, lower quartile and upper quartile.

Step 3: Draw a number line that will include the smallest and the largest data.

Step 4: Draw three vertical lines at the lower quartile (12), median (22) and the upper quartile (36), just above the number line.

Step 5: Join the lines for the lower quartile and the upper quartile to form a box.

Step 6: Draw a line from the smallest value (5) to the left side of the box and draw a line from the right side of the box to the biggest value (53).

Interquartile range

The interquartile range is the difference between the upper quartile and the lower quartile.

$$53 - 5 = 48$$

1. What are outlier?

Unusual data values as compared to the rest of the set. They may be distinguished by gaps in a histogram.

What are shapes of distribution?

Symmetric • Data is symmetric if the left half of its histogram is roughly a mirror image of its right half. Skewed • Data is skewed if it is not symmetric and if it extends more to one side than the other. Uniform • Data is uniform if it is equally distributed (on a histogram, all the bars are the same height or approximately the same height).

1. What are outlier?

Unusual data values as compared to the rest of the set. They may be distinguished by gaps in a histogram.

2. Name of probability sampling design?

Simple random sampling Stratified sampling? Systematic sampling Cluster (area) sampling multistage sampling?

3. Difference between qualitative quantitative variables?

Quantitative Variables It can be measured in

the usual sense. For example: - the heights of adult males, the weights of preschool

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children, the ages of patients seen in a - dental clinic. Qualitative Variables Many

characteristics are not capable of being measured. Some of them can be ordered or

ranked. For example: - classification of people into socioeconomic groups, -social classes based on income, education, etc

4. What are shapes of distribution?

Symmetric

Data is symmetric if the left half of its histogram is roughly a mirror image of its right half.

Skewed

- Data is skewed if it is not symmetric and if it extends more to one side than the other.

Uniform

- Data is uniform if it is equally distributed (on a histogram, all the bars are the same height or approximately the same height).

5. What is data and what are difference source

of data?

The raw material of Statistics is data. We may define data as figures. Figures result from the process of counting or from taking a measurement.

For example:

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When a hospital administrator counts the number of patients (counting). When a nurse weighs a patient (measurement) Sources: Routinely kept records, External sources. Surveys: Experiments

6. What is frequency distribution and

frequency histogram?

Frequency Distribution

- A table that organizes data values into classes or intervals along with number of values that fall in each class (frequency, f).

1. Ungrouped Frequency Distribution – for data

sets with few different values. Each value is in its own class.

2. Grouped Frequency Distribution:

For data sets with many different values, which are grouped together in the classes.

Frequency Histogram

A bar graph that represents the frequency distribution.

- The horizontal **scale** is quantitative and measures the data values.
- **The** vertical scale measures the frequencies of the classes.
- Consecutive bars must touch.

7. Write a comprehensive note on sampling and statistical inference. What are different types of sampling? What are the characteristics of each type of sampling?

Answer:

Sampling

A sample is a subset of individual from a larger population. Sampling means selecting the group that you will actually collect data from in your research.

For example, if you are researching the opinion of students in your university, you could survey a sample of 100 students.

Samples are used to make inference

about populations. Samples are easier to collect data from because they are practical, cost-effective, convenient and manageable.

Statistical inference

Statistical is the process of using data analysis to deduce properties of an underlying distribution of probability. Inferential statistics analysis infers properties of a population, for example by testing hypotheses and deriving estimates. It is assumed

that the observed data set is sampled from a

larger population.

Types of Sampling

sampling:

There are two major types of

Probability Sampling and

Non-probability sampling

Probability Sampling

Probability Sampling means

that every member of the target population has a known chance of being include in the sample.

Probability sampling method includes simple random sampling, systematic sampling, stratified sampling and cluster sampling.....

Random Sampling

Random sampling is analogous to putting everyone's name into a hat and drawing out several names. Each element in the population has an equal chance of occurring. While this is

the preferred way of sampling, it is often difficult to do. It requires that a complete list of element in the population be obtained.

Systematic Sampling

Systematic sampling is **easier** to do than random sampling. In systematic sampling, the list of element is "counted off". That is, every kith element is taken. This is similar to lining everyone up and numbering off "1, 2, 3, and 4; 1, 2, 3, **4**; etc". When done numbering, all people numbered 4 would be used.

Cluster Sampling

the

Cluster sampling is accomplished by dividing

population into groups_ usually geographically. These groups are called clusters or blocks. The clusters are randomly selected,

and each element in the selected clusters is

used.

Stratified Sampling

Stratified also divide the populations into groups called strata. However, this time it is by some characteristic, not geographically. For

instance, the population might be separated into males and females. A sample is taken from each of these strata using either random, systematic, or convenience.

NON-probability Sampling

Non probability

sampling is a sampling technique where the odds of any member being selected for a sample cannot be calculated. Example of non-probability sampling is convenience sampling, accidental sampling etc.

Convenience Sampling

Convenience sampling is very easy to do, but it is probably the worst technique to use. In convenience sampling readily available data is used. That is the first people the surveyor runs into.

Main characteristics of Sampling

In sampling, we assumed that samples are drawn from the population and sample means and population means are equal. A population can be defined as a whole that include all item and characteristics of the research taken into study. We make inference about the population with the help of sample.

Difference Qualitative and Quantitative

variable?

Variables can be classified as qualitative (aka, categorical) or quantitative (aka, numeric)

What is data how it can be collected? / What is

data and are different sources of data? What is

data enlist is sources?

A census is a study that obtains data from every member of a population. In most studies, a census is not practical, because of the cost and/or time required. Sample survey. A sample survey is a study that obtains data from a subset of a population, in order to estimate population attributes.

Write the different shapes of distribution? The

shape of a distribution is described by its

number of peaks and by its possession of symmetry, its tendency to skew, or

or its

uniformity. (Distributions that are skewed have more points plotted on one side of the graph than on the other.)

Non probability?

Non probability sampling which involves the sample being drawn from that part of the population which is close to hand. That is, readily available and convenient.

Frequency distribution and Histogram? Frequency distribution:

A table that organizes data values into classes

or intervals along with number of values that fall in each class (frequency, f).

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1. Ungrouped Frequency Distribution — for data

sets with few different values. Each value is in

its own class.

2. Grouped Frequency Distribution: for data sets with many different values, which are grouped together in the classes.

Frequency Histogram:

A bar graph that represents the frequency distribution.

- The horizontal scale is quantitative and measures the data values.
- The vertical scale measures the frequencies of the classes.
- Consecutive bars must touch.

Three types of sampling?

The main types of probability sampling methods

are simple random random sampling,
stratified sampling, cluster sampling,
multistage sampling, and systematic random sampling.

The key benefit of probability sampling methods is that they guarantee that the sample chosen is representative of the population.

What are outlier? / What do you mean by outlier?

An outlier is an observation point that is distant

from other observations.

An outlier may be due to variability in the measurement or it may indicate experimental error; the latter are sometimes excluded from the data set.

Name of probability sampling? Enlist type of probability sampling?

Simple Random Sampling. Simple random sampling is the easiest form of probability sampling. Stratified Random Sampling. Stratified random sampling is also known as proportional random sampling. Systematic Random Sampling.

Define biostatistics:

The use of statistical methods to solve the biological problems. It is also called biometry.

Population and Sample:

Population refers to the entire group of individuals that share a common characteristic and are the focus of a study at a particular time. In population the measurable quantity is called parameter.

While Sample is a subset of the population that is selected for the actual analysis. Sampling is often done when it is impossible to study the entire population. In sample the measurable quantity is called statistic.

Finite Population (Countable):

- The finite population is also known as a countable population in which the population can be counted.
- For statistical analysis, the finite population is more advantageous.
- Example: employees of a company.

Infinite Population (Uncountable):

- The population in which the counting of units in the population is not possible.
- It is less advantageous in statistical analysis.
- Example: Number of germs in the patient's body.

Parameter:

A parameter is a numerical characteristic of a population. It is a fixed value that describes some aspect of the population. For example, the average blood pressure of all adults in a specific region is a parameter.

Statistic:

A statistic is a numerical characteristic of a sample. It is used to estimate or infer the corresponding parameter of the population. For instance, the average blood pressure of a sample of 100 adults from the same region is a statistic.

Data:

Data refers to information collected, observed, or measured from population or sample during the study. This information can take various forms, including numerical values,

categorical labels, or descriptive details, and it serves as the foundation for statistical analysis. Biostatisticians use data to draw conclusions and derive meaningful insights about biological phenomena, health outcomes, or medical interventions.

There are two types of data:

1. Primary Data:

- This refers to data that is collected firsthand directly from the source.
- Common methods of collecting primary data include surveys, interviews, experiments, and observations.
- Primary data is original, providing a high degree of relevance and specificity to the research question.
- It is expensive and take more time.

2. Secondary Data:

- The that has already been collected by someone else for a different purpose.
- Researchers utilize existing sources, such as published studies, government reports, or datasets, to obtain secondary data.
- Secondary data is less specific to the researcher's needs and may not perfectly align with the study objectives.
- It is not expensive and time-saving way to access a large volume of information.

Variable:

A variable is a characteristic that can be measured and that can assume different values.

There are two types of variable:

1. Qualitative variable:

Qualitative variables are non-numeric and describe qualities or characteristics. Examples include gender, blood type, place of birth, ethnic group, type of drug etc.

2. Quantitative variable:

A quantitative variable is one that can be measured and expressed numerically. They have further two types:

- a) **Discrete Variable:** Discrete variables are numeric variables that can only take specific, distinct values, often counted in whole numbers and cannot be subdivided into smaller units. Examples include the number of patients in a clinic, the number of red blood cells.
- b) **Continuous Variable:** Continuous variables are numeric variables that can take any value within a range and can be measured with a high level of precision. These variables can be further subdivided into smaller and smaller units, and they are often associated with measurements rather than counts. Examples include height, weight, blood pressure, or temperature.

Sampling:

The process of selecting a subset from a larger population for study. There are two types:

1. **Probability sampling:**

In this type random selection take place and every element of the population has equal chance to be the part of selected sample. There many ways to do this: **a) Simple Random Sampling:**

Every individual in the population has an equal chance of being selected, and each selection is independent of others.

b) Stratified Random Sampling:

The population is divided into subgroups or strata based on certain characteristics, and then random samples are taken from each stratum.

c) Cluster Sampling:

The population is divided into clusters, and then a random sample of clusters is selected. All members within the chosen clusters are included in the sample.

- Stratified sampling, which involves selecting from every subgroup, cluster sampling involves selecting entire groups.

d) Systematic Sampling:

- A random starting point is selected, and then every k th element in the population is chosen for the sample.

e) Multistage Sampling:

Involves multiple stages of sampling. It often combines various sampling methods, such as cluster sampling followed by simple random sampling within selected clusters.

2. Non- Probability Sampling:

The sampling technique where not every member of the population has an equal chance of being included in the sample.

Some common types of non-probability sampling methods include:

a) Convenience Sampling:

Participants are selected based on availability.

b) Purposive Sampling:

Choose based on the purpose of the study.

c) Quota Sampling:

Choose on the base of some pre-set standard. Proportion of the sample should be same as population.

d) Snowball Sampling:

First participants give the future participants. An initial set of participants is selected through purposeful sampling. These participants are then asked to refer others who might meet the study criteria.

Mean:

The mean, or average, is calculated by summing up all the values in a dataset and dividing the sum by the number of values.

Example:

Consider the dataset {12, 15, 18, 22, 25}.

$$\text{Mean} = (12 + 15 + 18 + 22 + 25) / 5 = 92 / 5 = \underline{18.4}$$

Mode:

The mode is the value that appears most frequently in a dataset.

Example:

For the dataset {5, 8, 12, 8, 15, 8, 22}, the mode is 8 because it appears more frequently than any other value.

Median:

The median is the middle value when a dataset is ordered. If the dataset has an even number of values, the median is the average of the two middle values.

Example:

For the dataset {10, 15, 18, 22, 25}, the median is 18. For the dataset {10, 15, 18, 22}, the median is $(15 + 18) / 2 = 16.5$.

Range:

The range is the difference between the maximum and minimum values in a dataset.

Example:

For the dataset {8, 12, 5, 15, 22}, the range is $22 \text{ (maximum)} - 5 \text{ (minimum)} = 17$

Branches/ Types of Biostatistics:**1. Inferential Statistics:**

Inferential statistics, on the other hand, go beyond the immediate data and use probability theory to make inferences or predictions about a population based on a sample of data. This includes techniques like hypothesis testing, confidence intervals, and regression analysis. Inferential statistics help draw conclusions and make generalizations about populations by analyzing a subset of the data.

2. Descriptive Statistics:

Descriptive statistics involve methods for summarizing and presenting data in a meaningful way. This includes measures of central tendency like the mean, median, and mode, as well as measures of dispersion like range and standard deviation. Descriptive statistics aim to provide a clear and concise overview of the main features of a dataset.

- Descriptive statistics is performed using **Tabular and Graphical Methods**.

i) **Tubular Method:** It is used to summarize the data in table form. The most frequent method is used in tabular form is frequency table.

ii) **Graphical Methods:** It is visual way of parenting data using charts and graphs. The most frequent methods are used for this purpose are Histogram, Box, Whisker Plot, Scatter plot etc.

Frequency Distribution:

Frequency refers to the number of times a particular value occurs in a dataset and Frequency Distribution is a method used to organize and summarize the data in tables and give more insight of the data.

There are three types of frequency distribution.

1. **Categorical Frequency Distribution:** It is used for qualitative data where categories have no any order such as colour, gender etc.

2. **Ungrouped Frequency Distribution:** It is used for the quantitative data mainly dealing with small dataset and each value is treated separately.

3. **Grouped Frequency Distribution:** Also used for quantitative data but deals with larger data set where data grouped into many interval or classes.

- **Relative Frequency** = frequency of the class / Total
- **Cumulative Relative Frequency** = sum of previous relative Frequencies + current class frequency.

In categorical and ungrouped frequency distribution we make a table. Write the data (avoid repetition) and use Tally Marks for counting and write frequency in 3rd column. We can also find relative frequency and Cumulative Relative Frequency in 4th and 5th columns using above formulas.

Steps for Grouped Frequency Distribution:

1. Write the data in order array.
2. Find the smallest value (X_o) and largest value (X_m)
3. Find the range which is difference between largest and smallest value.
4. Find width (class interval) denoted by h : divide range by number of chosen classes (Often 5 to 15 classes to be assumed).
5. Make number of classes from smallest value to largest value based on class interval.
6. Now draw tally mark and find the frequency of the dataset.

Frequency Histogram:

After the above process if we make bars for visual display of the data then it is called frequency histogram. The height of the bar represent the frequency.

Best video for frequency distribution explanation:

<https://youtu.be/riYpfCluOUo?si=EsjjuDC4qWKHF5ek>

Box and Whisker Plot Method:

This method is used for graphical presentation of data.

Step:

1. Arrange the data in ascending order.
2. Find mean (50% of data), lower quartile (25%), upper quartile (75%) of the data.
3. Draw a number line that include smallest and largest value of the data.
4. Draw three vertical lines at the lower quartile, mean and upper quartile above the number line.
5. Join the lines for lower quartile and upper quartile to form a box
6. Draw the line from box at left side for smallest value and a line from box to right for largest value.
7. The difference between upper and lower quartile is called interquartile range (IQR).

Video explanation:

<https://youtu.be/7jP3TowC9U4?si=4BJoYzmpV5zHt7KH>

Permutations:

The arrangements of objects in a specific order.

Case 1:

When all objects are distinct:

Formula: $nPr = \frac{n!}{(n-r)!}$

Example:

A club consists of four members. How many ways are there of selecting three officers: president, secretary and treasurer? It is evident that the order, in which 3 officers are to be chosen, is of significance.

n
=
4
r
=
3

Put in formula:

$$4P3 = \frac{4!}{(4-3)!}$$

$$4 \times 3 \times 2 \times 1 = 24$$

Case 2:

When some objects are repeated:

Formula:

$$P = \frac{n!}{n_1! \times n_2! \times n_3! \dots}$$

Example:

Find number of permutations in the word "BOOKKEEPER,"

- Total number of letter (n) = 10

- O = 2

- K = 2

- E = 3

$$P = 10! / 2! \times 2! \times 3!$$

$$P = 151200$$

Combination :

The selections of elements without considering the order.

Formula

:

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

:

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Example:

A group of 5 people, and you want to form a committee of 2 people. Find number of combination.

n

=

5

r

=

2

$$C = 5! / 2! (5-2) !$$

$$C = 10 \text{ combinations}$$

Video Explanation:

Short video : https://youtu.be/GZATQeI8dJo?si=H_IXfmUA7GrShg6y

Long video complete concept: <https://youtu.be/Tr-TVt5JAWY?si=3Ozrz9wWSERRsnxZ>

Mean Deviation:

Example : Set of data: [2, 4, 6, 8, 10]. Find mean deviation.

1. Calculate the Mean:

$$2 + 4 + 6 + 8 + 10 / 5 = 6$$

2. Find the Deviation for Each Data Point:

- Deviation for 2: $(2 - 6 = -4)$
- Deviation for 4: $(4 - 6 = -2)$
- Deviation for 6: $(6 - 6 = 0)$
- Deviation for 8: $(8 - 6 = 2)$
- Deviation for 10: $(10 - 6 = 4)$

3. Sum of Deviations:

$$[(-4) + (-2) + 0 + 2 + 4 = 0]$$

Variance:

1. Square the mean deviation and then sum up and divide by number of dataset. $16 + 4 + 0 + 4 + 16 / 5 = 8$

Standard Deviation:

Take the square root of variance value

$$\sqrt{8} = 2.83$$

Coefficient of standard deviation = Standard deviation / Mean x

$$100 \cdot 2.83 / 6 \times 100 = 47.16$$

Video Explanation:

https://youtu.be/tXv1KVoQrWA?si=y_ZS0-JQSc62x5d

<https://youtu.be/179ce7ZzFA8?si=77AQC7y8qFUGmlqr>

Probability = Number of favorable outcomes/ Total number of possible outcomes

Example:

Tossing a fair six-sided die. Find the Probability of Rolling an Even Number:

Possible outcomes = 2,4,6

Total Sides = 6

$P = 3/6 = 1/3$

Video Explanation: <https://youtube.com/shorts/6oCvo2BULbU?si=JEx8UtCxynvw889l>

Q.1 What is Random error?

An error is said to be random when the observed value is consistently and constantly higher or lower than the true value. Biased errors arise from the personal limitations of the observer, the imperfection in the instruments used or some other conditions which control the measurements.

Q. What are types of sampling ?

There are two types of sampling methods:

- Probability sampling involves random selection, allowing you to make strong statistical inferences about the whole group.
- Simple random sampling is a completely random method of selecting subjects. These can include assigning numbers to all subjects and then using a random number generator to choose random numbers. The members whose numbers are chosen are included in the sample.
- Stratified Random Sampling involves splitting subjects into mutually exclusive groups and then using simple random sampling to choose members from groups.
- Systematic Sampling means that you choose every “nth” participant from a complete list. For example, you could choose every 10th person listed.
- Cluster Random Sampling is a way to randomly select participants from a list that is too large for simple random sampling. For example, if you wanted to choose 1000 participants from the entire population of the U.S., it is likely impossible to get a

complete list of everyone. Instead, the researcher randomly selects areas (i.e. cities or counties) and randomly selects from within those boundaries.

- Multi-Stage Random sampling uses a combination of techniques.
- Non-probability sampling involves non-random selection based on convenience or other criteria, allowing you to easily collect data.
- Convenience sampling is a non-probability sampling technique where samples are selected from the population only because they are conveniently available to the researcher. Researchers choose these samples just because they are easy to recruit, and the researcher did not consider selecting a sample that represents the entire population.

- Consecutive sampling:

This non-probability sampling method is very similar to convenience sampling, with a slight variation. Here, the researcher picks a single person or a group of a sample, conducts research over a period, analyzes the results, and then moves on to another subject or group if needed. Consecutive sampling technique gives the researcher a chance to work with many topics and fine-tune his/her research by collecting results that have vital insights.

- Quota sampling:

Hypothetically consider, a researcher wants to study the career goals of male and female employees in an organization. There are 500 employees in the organization, also known as the population. To understand better about a population, the researcher will need only a sample, not the entire population. Further, the researcher is interested in particular strata within the population. Here is where quota sampling helps in dividing the population into strata or groups.

- Judgmental or Purposive sampling:

In the judgmental sampling method, researchers select the samples based purely on the researcher's knowledge and credibility. In other words, researchers choose only those people who they deem fit to participate in the research study. Judgmental or purposive sampling is not a scientific method of sampling, and the downside to this sampling technique is that the preconceived notions of a researcher can influence the results. Thus, this research technique involves a high amount of ambiguity.

- Snowball sampling:

Snowball sampling helps researchers find a sample when they are difficult to locate. Researchers use this technique when the sample size is small and not easily available. This sampling system works like the referral program. Once the researchers find suitable subjects, he asks them for assistance to seek similar subjects to form a considerably good size sample.

Q.2 What is Parameters and Frame?

In order to draw a random sample from a finite population, the first thing that we need is the complete list of all the elements in our population. This list is technically called the FRAME.

In statistics, a parameter is any measured quantity of a statistical population that summarises or describes an aspect of the population, such as a mean or a standard deviation.

Q.3 Define Quantitative variable & Qualitative variable.

A variable is called a *quantitative variable* when a characteristic can be expressed numerically such as age, weight, income or number of children. On the other hand, if the characteristic is non-numerical such as education, sex, eye-colour, quality, intelligence, poverty, satisfaction, etc. the variable is referred to as a *qualitative variable*. A qualitative characteristic is also called an *attribute*. An individual or an object with such a characteristic can be counted or enumerated after having been assigned to one of the several mutually exclusive classes or categories.

Q.4 What is Window Variable editor?

The data editor window is the default window when you run SPSS. The data worksheet works just like a spreadsheet, where a column represents a variable and a row represents a case or an observation.

Q.5 When Skewness was discovered?

Between the end of the nineteenth century and the beginning of the twentieth century, Pearson, Karl studied large sets of data which sometimes deviated significantly from normality and exhibited considerable skewness.

Q.6 What is Dispersion?

In statistics, dispersion is the extent to which a distribution is stretched or squeezed. Common examples of measures of statistical dispersion are the variance, standard deviation, and interquartile range. For instance, when the variance of data in a set is large, the data is widely scattered.

Q.7 What is Relative measure of dispersion?

Relative measures of dispersion are measures of the variance of a range of values regardless of its unit of measure. This means that the spread of two ranges of values with different measures can be compared directly with relative measures of dispersion.

Q.8 What is Regression analysis?

Regression analysis is a powerful statistical method that allows you to examine the relationship between two or more variables of interest. While there are many types of

regression analysis, at their core they all examine the influence of one or more independent variables on a dependent variable.

Q.9 What is Data scale?

Scales of measurement in research and statistics are the different ways in which variables are defined and grouped into different categories. Sometimes called the level of measurement, it describes the nature of the values assigned to the variables in a data set.

There are four types of data scale:

- i. Nominal
- ii. Ordinal
- iii. Interval
- iv. Ratio

Q.10 What is the difference between quota and cluster sampling?

Quota Sampling:-

In this type of sampling, the selection of the sampling unit from the population is no longer dictated by chance. A sampling frame is not used at all, and the choice of the actual sample units to be interviewed is left to the discretion of the interviewer. However, the interviewer is restricted by quota controls.

Cluster Sampling:-

Cluster sampling is a probability sampling method in which you divide a population into clusters, such as districts or schools, and then randomly select some of these clusters as your sample. The clusters should ideally each be mini-representations of the population as a whole.

Q.11 Explain Range rule of thumb.

The range rule of thumb is a handy method of estimating the range from the standard deviation. It tells us that the range is generally about four times the standard deviation. So if your standard deviation is 2, you might guess that your range is about eight.

Q.12 Write general and specific rule for dependent and independent events.

Independent Events:

Two events are independent if the occurrence of one does not change the probability of the other occurring. An example would be rolling a 2 on a dice and flipping a head on a coin. Rolling the 2 does not affect the probability of flipping the head.

Dependent Events:

Two events are dependent when the outcome of the first event influences the outcome of the second event. The probability of two dependent events is the product of the probability of X and the probability of Y AFTER X occurs.

Q.13 What are Outliers?

An outlier is an observation that lies an abnormal distance from other values in a random sample from a population.

A value that "lies outside" (is much smaller or larger than) most of the other values in a set of data. For example in the scores 25,29,3,32,85,33,27,28 both 3 and 85 are "outliers".

Q.14 What are Distribution types?

There are two types of probability distribution which are used for different purposes and various types of the data generation process. Normal or Cumulative Probability Distribution. Binomial or Discrete Probability Distribution.

Q.15 Define Correlation.

CORRELATION is a measure of the strength or the degree of relationship between two RANDOM variables.

A numerical measure of the strength of the linear relationship between two random variables X and Y is known as Pearson's Product-Moment Coefficient of Correlation. Correlation can be linear, positive, perfect positive, negative, perfect negative and there can be no correlation at all.

Q.16 How many shapes of distribution are there?

Generally there are two basic types of distribution. Distribution may be symmetric or asymmetric but in advanced statistics, there are seven types of distribution. They are:

- i. Symmetric
- ii. Bell shaped
- iii. U shaped
- iv. Skewed
- v. Positively skewed
- vi. Negatively skewed
- vii. Uniform

Q.17 Name types of probability sampling.

Simple random sampling

Stratified Random Sampling

Systematic Sampling

Cluster Random Sampling

Multi-Stage Random sampling

Q.18 What is Regression equation?

A regression equation is used in stats to find out what relationship, if any, exists between sets of data. For example, if you measure a child's height every year you might find that they grow about 3 inches a year. That trend (growing three inches a year) can be modeled with a regression equation.

$$Y_i = f(X_i, \beta) + e_i$$

Y_i = dependent variable
 f = function
 X_i = independent variable
 B = unknown parameters
 e_i = error terms

Q.19 What are categories of diagnostic test?

There are many different types of tests in statistics like t-test, Z-test, chi-square test, anova test, binomial test, one sample median test etc. Parametric tests are used if the data is normally distributed.

Q.20 Write field names where statistics is used.

Not only in the field of natural and physical science but statistics has made much impact on social sciences and is of immense use in these also. These include subjects like History, Political Science, Education, Psychology, Sociology and Logic etc. Even in mathematics, many theories have been drawn from statistics.

Q.21 What is Data and its resources?

Data is basically divided into two categories.

- i. Primary Data
- ii. Secondary Data

Primary data can further be divided into:

- a. Interview method
- b. Survey method
- c. Observation method
- d. Experimental method

Secondary data can further be classified as:

- a. Internal source
- b. External source

Q.22 What is primary and secondary data?

PRIMARY DATA

Primary data is a type of data that is collected by researchers directly from main sources through interviews, surveys, experiments, etc. Primary data are usually collected from the source—where the data originally originates from and are regarded as the best kind of data in research.

SECONDARY DATA

Secondary data is the data that has already been collected through primary sources and made readily available for researchers to use for their own research. It is a type of data that has already been collected in the past.

A researcher may have collected the data for a particular project, then made it available to be used by another researcher. The data may also have been collected for general use with no specific research purpose like in the case of the national census.

Q.23 What is Percentile?

In statistics, a k-th percentile is a score below which a given percentage k of scores in its frequency distribution falls or a score at or below which a given percentage falls. For example, the 50th percentile is the score below which or at or below which 50% of the scores in the distribution may be found.

Q.24 What are different types of graphs?

Types of Graphs and Charts

- Bar Chart/Graph.
- Pie Chart.
- Line Graph or Chart.
- Histogram Chart.
- Area Chart.
- Dot Graph or Plot.
- Scatter Plot.
- Bubble Chart.

Q.25 What is regression and correlation?

Regression is a statistical method used in finance, investing, and other disciplines that attempts to determine the strength and character of the relationship between one dependent variable (usually denoted by Y) and a series of other variables (known as independent variables).

Correlation is a statistical measure that expresses the extent to which two variables are linearly related (meaning they change together at a constant rate). It's a common tool for describing simple relationships without making a statement about cause and effect.

Q.26 Define Discrete and Continuous variable.

A discrete variable is a variable whose value is obtained by counting. Examples: number of students present. number of red marbles in a jar. number of heads when flipping three coins.

Continuous variables are numeric variables that have an infinite number of values between any two values. A continuous variable can be numeric or date/time. For example, the length of a part or the date and time a payment is received.

Q.27 What is pie chart?

A pie chart, sometimes called a circle chart, is a way of summarizing a set of nominal data or displaying the different values of a given variable (e.g. percentage distribution). This type of chart is a circle divided into a series of segments. Each segment represents a particular category.

Q.28 How to make box and whisker plot?

In a box plot, we draw a box from the first quartile to the third quartile. A vertical line goes through the box at the median. The whiskers go from each quartile to the minimum or maximum.

Q.29 Write two regression equation.

The equation has the form $Y = a + bX$, where Y is the dependent variable (that's the variable that goes on the Y axis), X is the independent variable (i.e. it is plotted on the X axis), b is the slope of the line and a is the y-intercept.